

Subhertz linewidth laser by locking to a fiber delay line

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An ultralow-noise, subhertz 1.55 μm erbium-doped fiber laser that is locked on an all-fiber-based Michelson interferometer is presented in this paper. The interferometer uses 500 m SMF-28 optical fiber and an acousto-optic modulator to allow heterodyne detection. By comparing two identical laser systems, a 0.67 (0.21) Hz linewidth beat-note signal is achieved and we obtain fractional frequency instability of 7×10^{-15} at short timescales (0.1–1 s). The frequency noise power spectral density of two identical lasers is below $-1 \text{ dB Hz}^2/\text{Hz}$ at 1 Hz and it reaches $-18 \text{ dB Hz}^2/\text{Hz}$ from 200 Hz to 1 kHz. © 2015 Optical Society of America

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1. Introduction

Ultranarrow-linewidth lasers are important tools for many applications. In the field of frequency metrology, ultranarrow-linewidth lasers are used as local oscillators of the new generation of optical atomic clocks [1–3], and for generation of ultralow phase noise microwave signals via femtosecond optical frequency combs [4]. Moreover, they play key roles in modern precision measurement, such as gravitational wave detection [5,6], precision testing of the theory of relativity [7], and high-resolution spectroscopy [8]. Currently, the lowest noise lasers are realized by stabilizing laser frequency onto an ultrastable high-finesse Fabry–Perot cavity with the Pound–Drever–Hall (PDH) method [9,10]. However, this approach requires stable and precise alignment of free-space optical elements, high vacuum, and even a cryostat. Therefore, the system is bulky, fragile, and expensive. A radical alternative is to use an optical fiber delay line as a frequency reference to stabilize laser frequency. Recently, Kéfélian *et al.* [11] and Jiang *et al.* [12] demonstrated that fiber-stabilized lasers show

comparable frequency noise to those obtained by PDH locking to a high-finesse cavity for a Fourier frequency from 40 Hz to 30 kHz and a frequency noise level below $1 \text{ Hz}^2/\text{Hz}$ at 1 Hz.

In this paper, we demonstrate a subhertz linewidth laser stabilized to an all-fiber-based interferometer. In contrast to previous works [11,12], we improve the environmental isolation by means of active temperature stabilization and by placing the interferometer into a low-level vacuum, which dramatically reduce the frequency drift and the noise at low Fourier frequency. Two fiber-stabilized laser systems are constructed and we can fully evaluate the performance of each. For two identical systems, we achieve a 0.67 (0.21) Hz linewidth beat-note signal and the frequency noise of $-1 \text{ dB Hz}^2/\text{Hz}$ at 1 Hz and $-18 \text{ dB Hz}^2/\text{Hz}$ from 200 Hz to 1 kHz.

2. Operation Principle and Experimental Setup

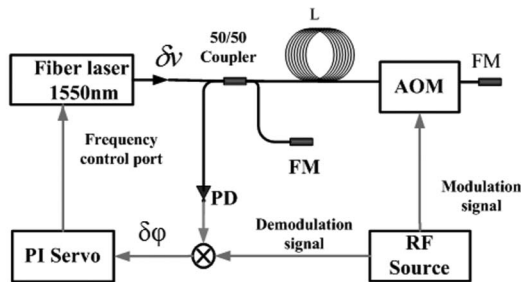
In Fig. 1(a), the heterodyne arm-unbalanced Michelson interferometer acts as a frequency discriminator to convert the frequency fluctuation ($\delta\nu$) of the laser into the phase fluctuation ($\delta\varphi$) of the heterodyne detected RF signal. The phase fluctuation is detected and fed to the laser to correct its frequency fluctuation. The transfer function [12] of the heterodyne

signal phase fluctuation $\delta\varphi(f)$ to the laser frequency fluctuation $\delta\nu(f)$ is

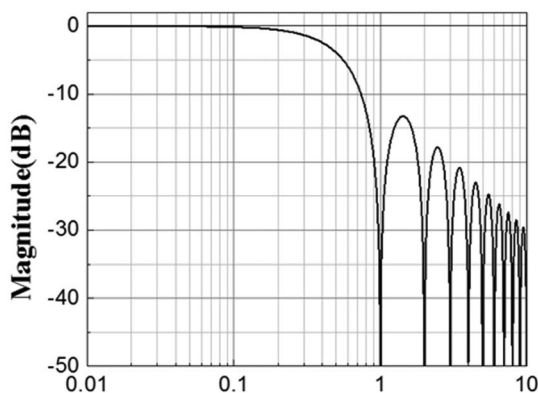
$$T(f) = \frac{\delta\varphi(f)}{\delta\nu(f)} = \frac{1 - e^{-i2\pi f\tau}}{if}, \text{ rad/Hz}, \quad (1)$$

where f is the Fourier frequency, τ is the unbalanced time delay of the interferometer $\tau = 2nL/c$, L is the fiber length difference between the two arms, n is the effective refractive index, and c is the speed of light in vacuum. This function is plotted for magnitude in Fig. 1(b); at low frequencies ($f \ll 1/\tau$), the magnitude is approximately equal to $2\pi\tau$, which means a longer fiber increases the gain of the discriminator while it decreases the bandwidth of the control loop.

Figure 2 shows the scheme of the laser frequency stabilization system. A 1.55 μm erbium-doped fiber laser is used as the laser source, the laser beam goes to a Michelson interferometer passing through an acousto-optic modulator (AOM1). Another acousto-optic modulator (AOM2) is inserted into the long arm of interferometer after a 500 m fiber spool and driven by a 75 MHz RF signal. In the interferometer, the Faraday mirrors guarantee maximum beat-note signal amplitude without requiring any polarization controller. The light power that injects into the interferometer is about 500 μW . At the photodiode, a 150 MHz RF signal is detected. We use a homemade



(a) Principle of laser stabilization by fiber delay line



(b) Fourier frequency(normalized by $1/\tau$)

Fig. 1. (a) Principle of laser stabilization by fiber delay line. (b) Transfer function of the interferometer (relative magnitude $T(f)$). AOM, acousto-optic modulator; FM, Faraday-mirror; PD, photodiode; PI, proportional-integral; RF, radio frequency.

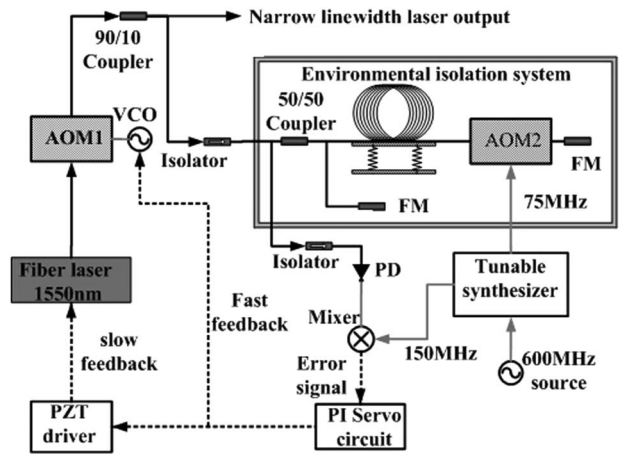


Fig. 2. Scheme of the laser frequency stabilization system. VCO, voltage-controlled oscillator; PZT, piezoelectric transducer. The optical signals, RF signals, and low-frequency signals are shown by the black solid, gray solid, and black dashed lines, respectively.

low phase noise tunable synthesizer to provide a demodulation signal and a PI servo circuit to convert the demodulated signal into a laser frequency correction signal. The correction signal simultaneously acts on a piezo-electric transducer (PZT) stretcher and a voltage controlled oscillator (VCO), which drives AOM1. The control of the PZT has a large dynamic range with a slow response, while the control of the AOM has a small dynamic range with a fast response. Consequently, we combine the benefits of both control ports to correct the laser frequency noise with a high control bandwidth and a large dynamic range.

In our experiment, the fiber spool is placed into an aluminum air-sealed can and a vacuum bag is used to contain the interferometer to reduce the influence of air-flow and temperature fluctuation. The whole interferometer system is placed on a passive vibration isolation platform (BM-4, Minus K Inc.) and housed by an aluminum box, which is covered with acoustic-damping foam and thermal insulation foam to isolate vibration noise, acoustic noise, and temperature fluctuation. The aluminum box is single point temperature stabilized at about 301 K by an active thermal controller. The total temperature fluctuation is below 10 mK over a 24 h period.

Different from traditional rigid cavity methods, the fiber-stabilized laser is constructed by all-fiber optical links, which provides a simple structure, occupies small volume, and requires no optical alignment. The total volume of the optical links is about 15 l and can be further minimized, making it smaller than most cavity stabilized lasers. Moreover, the fiber-stabilized laser can be used as an agile laser with a wide frequency tuning range when a frequency offset is applied to the tunable synthesizer [12].

3. Measurement Results and Discussion

We set up two fiber-stabilized lasers for comparison. A second fiber-stabilized laser was constructed with a separate fiber laser, separate electronics, and a

separate all-fiber-based Michelson interferometer mounted on an independent vibration isolation platform and housed in an independent enclosure. However, the two systems are identical and their contributions to the measurements can be considered the same.

The RF beat-note signal (about 1.3 GHz) between two fiber-stabilized lasers is analyzed by a phase noise measurement system (N5500A, Agilent) to obtain its phase noise, and frequency noise $S_{\phi}^*(f)$ can be derived from phase noise $S_{\phi}(f)$ by

$$S_{\phi}^*(f) = f^2 S_{\phi}(f), \quad (2)$$

where f is the Fourier frequency. The result is presented in Fig. 3. For Fourier frequencies lower than 1 kHz, the frequency noise of fiber-stabilized lasers is reduced more than 47 dB compared to that of free-running lasers. At high frequency, the noise increases due to the limited gain of the servo loop.

Figure 4 plots the measured frequency noise of fiber-stabilized lasers together with noise sources in our experiment. The noise floor of the measurement system [curve (b)] and the tunable synthesizer [curve (c)] is more than 20 dB lower than the measured frequency noise of fiber-stabilized lasers, so these influences can be neglected. The measured frequency noise [curve (a)] is less than $0.8 \text{ Hz}^2/\text{Hz}$ ($-1 \text{ dB Hz}^2/\text{Hz}$) at 1 Hz and it achieved about $0.016 \text{ Hz}^2/\text{Hz}$ ($-18 \text{ dB Hz}^2/\text{Hz}$) from 200 Hz to 1 kHz. Note that, for one fiber-stabilized laser, the result should be subtracted by 3 dB, which is slightly better than that of the reports in [11,12], which use 1 and 2.5 km fiber spools, respectively. Curve (d) is a theoretical simulation of the thermal noise limit for two 1 km length fibers (in a Michelson scheme, light passes through the fiber spool twice) according to Wanser's theory [13]. However, this theoretical result has good agreement with experimental results only at high Fourier frequency [14,15]. Curve (e) is the thermal noise from the first-stage RF amplifier after the photodiode. In our experiment, obviously,

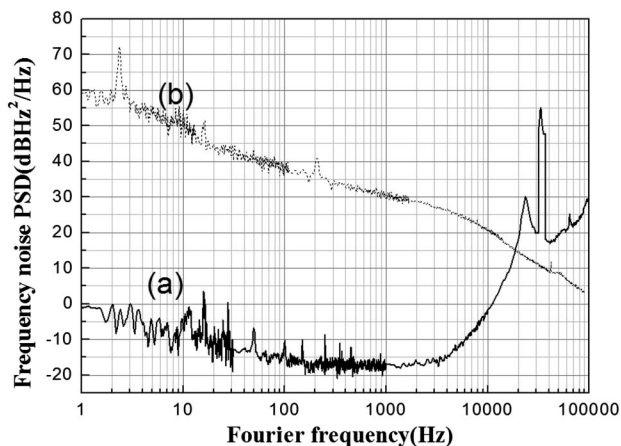


Fig. 3. Frequency noise PSD of (a) two identical fiber-stabilized lasers (solid line) and (b) two free-running fiber lasers (dotted line).

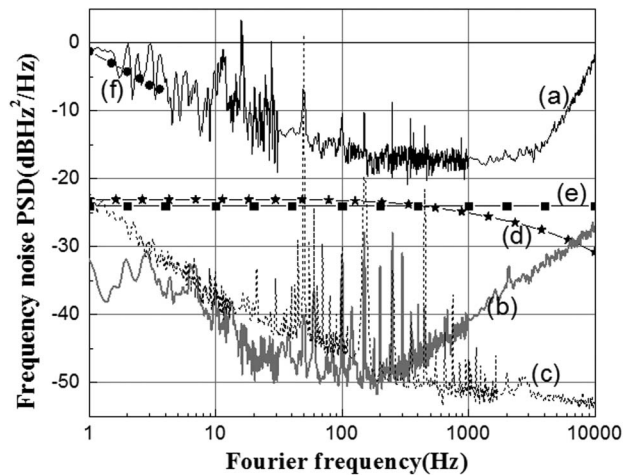


Fig. 4. Frequency noise PSD of (a) two identical fiber-stabilized lasers (black solid), (b) noise floor of the measurement system (gray solid), (c) noise floor of the tunable synthesizer (black dotted), (d) intrinsic thermal noise floor according to Wanser's theory (stars), (e) thermal noise of the first-stage amplifier after the photodiode (squares), and (f) intrinsic thermal noise due to mechanical dissipation, for Fourier frequency lower than 3.6 Hz (circles).

the frequency noise from 200 Hz to 1 kHz is dominated by the combination of these two noise sources and has only about 2 dB discrepancy with the measured result. Recently, a new calculation of spontaneous length fluctuation of optical fiber caused by mechanical dissipation was done by Duan [16]. This approach use a one-dimensional model based on the fluctuation-dissipation theorem (FDT) and is valid only when the Fourier frequency f is lower than $1/\tau_l$, where τ_l is the time it takes for the sound wave to travel across the length of the fiber. For our experiment, $1/\tau_l$ is equal to 3.6 Hz. The calculated thermal noise floor based on Duan's method is presented as

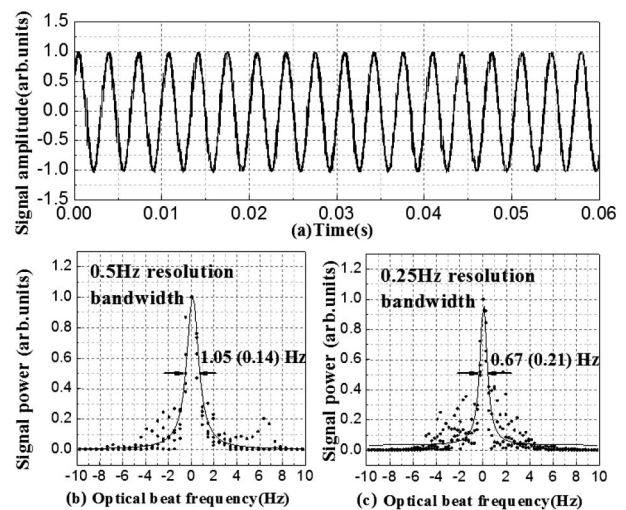


Fig. 5. (a) Heterodyne beat-note signal of two stabilized lasers and laser linewidth measurements at (b) 0.5 Hz resolution bandwidth and (c) 0.25 Hz resolution bandwidth. The Lorentzian fits are indicated in (b) and (c) by solid lines. The combined result of five (0.5 Hz RBW) and four (0.25 Hz RBW) consecutive recordings of the beat signal (black dots) are displayed, respectively.

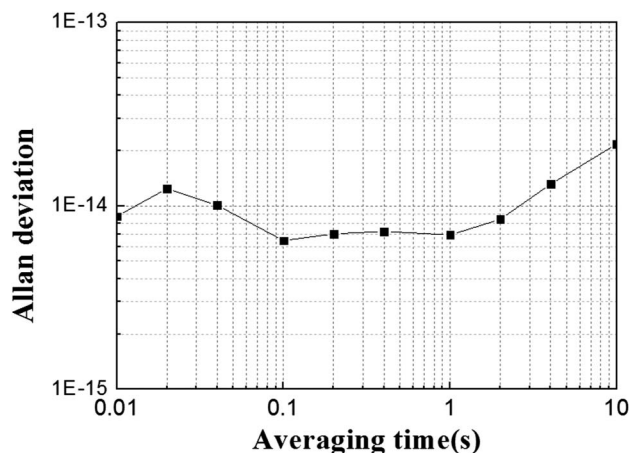


Fig. 6. Fractional frequency instability of two fiber-stabilized lasers.

curve (f) in Fig. 4. Below 3.6 Hz, the curve presents $1/f$ dependence and fits the measured result very well, which means the laser noise is dominated by intrinsic thermal noise. However, the experimental result range from 3.6 to 100 Hz exhibits a $1/f$ characteristic seems also consistent with Duan's theory but exceeds the frequency restrictions of the theory. This contradiction requires further investigation in future work.

To measure the laser linewidth, the heterodyne beat-note signal (about 1.3 GHz) between two fiber-stabilized lasers was mixed with a RF signal from a signal generator (E8257D, Agilent) to obtain a low-frequency signal [~ 300 Hz; see Fig. 5(a)]. The linear drifts were removed by applying a very small frequency offset on the tunable synthesizer. Thanks to the active temperature control and thermal isolation, the drift was reduced to about 1–2 Hz/s. Then, this signal was analyzed by a fast Fourier transform spectrum analyzer (SR760, Stanford Research Systems). At 0.5 Hz resolution bandwidth (RBW), the laser linewidth is 1.05 (0.14) Hz [full width at half-maximum, Fig. 5(b)]. As the RBW is reduced to 0.25 Hz, a linewidth of 0.67 (0.21) Hz can be observed [Fig. 5(c)].

For fractional frequency instability measurement, the heterodyne beat-note signal between two fiber-stabilized lasers under linear drift cancellation was tuned to 640 MHz and divided by 128 using a low noise frequency divider. The obtained 5 MHz signal was compared with a high-performance oven controlled crystal oscillator (BVA8600, Oscilloquartz SA) via an Allan deviation test set (TSC5110A, Timing Solutions). The measured results are shown in Fig. 6. We achieve fractional frequency instability of 7×10^{-15} at short timescales 0.1–1 s (without removing the frequency drift). This means that, for one fiber-stabilized laser, the fractional frequency instability is below 5×10^{-15} (0.1–1 s).

4. Conclusions

In conclusion, two reliable, simple, low-cost, and frequency tunable subhertz fiber-stabilized lasers were developed. By comparing the two lasers, we obtain a

0.67 (0.21) Hz linewidth beat-note signal and measure the frequency noise and the frequency instability of the two lasers. The frequency noise is below -1 dB Hz²/Hz at 1 Hz and reaches -18 dB Hz²/Hz from 200 Hz to 1 kHz. The fractional frequency instability shows a level of 7×10^{-15} from 0.1 to 1 s. In the future, our work should focus on improving the stability of the fiber-stabilized laser and further minimizing the system. These can be accomplished three ways: (1) use longer fiber to reduce thermal noise limit, (2) put the interferometer into a vacuum tank with several thermal shields to reduce the influence of temperature fluctuation, and (3) use a low vibration sensitivity fiber spool instead of a passive vibration isolation platform to minimize system size and complexity.

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References

1. C. W. Chou, D. B. Hume, J. C. J. Koelemeij, D. J. Wineland, and T. Rosenband, "Frequency comparison of two high-accuracy Al⁺ optical clocks," *Phys. Rev. Lett.* **104**, 070802 (2010).
2. A. D. Ludlow, T. Zelevinsky, G. K. Campbell, S. Blatt, M. M. Boyd, M. H. G. de Miranda, M. J. Martin, J. W. Thomsen, S. M. Foreman, J. Ye, T. M. Fortier, J. E. Stalnaker, S. A. Diddams, Y. Le Coq, Z. W. Barber, N. Poli, N. D. Lemke, K. M. Beck, and C. W. Oates, "Sr lattice clock at 1×10^{-16} fractional uncertainty by remote optical evaluation with a Ca clock," *Science* **319**, 1805–1808 (2008).
3. Chr. Tamm, S. Weyers, B. Lipphardt, and E. Peik, "Stray-field induced quadrupole shift and absolute frequency of the 688-THz $^{171}\text{Yb}^+$ single-ion optical frequency standard," *Phys. Rev. A* **80**, 043403 (2009).
4. T. M. Fortier, M. S. Kirchner, F. Quinlan, J. Taylor, J. C. Bergquist, T. Rosenband, N. Lemke, A. Ludlow, Y. Jiang, C. W. Oates, and S. A. Diddams, "Generation of ultrastable microwaves via optical frequency division," *Nat. Photonics* **5**, 425–429 (2011).
5. G. M. Harry, H. Armandula, E. Black, D. R. M. Crooks, G. Cagnoli, J. Hough, P. Murray, S. Reid, S. Rowan, P. Sneddon, M. M. Fejer, R. Route, and S. D. Penn, "Thermal noise from optical coatings in gravitational wave detectors," *Appl. Opt.* **45**, 1569–1574 (2006).
6. The LIGO Scientific Collaboration, "LIGO: the laser interferometer gravitational-wave observatory," *Rep. Prog. Phys.* **72**, 076901 (2009).
7. Ch. Eisele, A. Yu. Nevsky, and S. Schiller, "Laboratory test of the isotropy of light propagation at the 10^{-17} level," *Phys. Rev. Lett.* **103**, 090401 (2009).
8. W. H. Oskay, W. M. Itano, and J. C. Bergquist, "Measurement of the $^{199}\text{Hg}^+ 5d^9 6s^2 2D_{5/2}$ electric quadrupole moment and a constraint on the quadrupole shift," *Phys. Rev. Lett.* **94**, 163001 (2005).
9. R. W. P. Drever, J. L. Hall, F. V. Kowalski, J. Hough, G. M. Ford, A. J. Munley, and H. Ward, "Laser phase and frequency stabilization using an optical resonator," *Appl. Phys. B* **31**, 97–105 (1983).
10. T. Kessler, C. Hagemann, C. Grebing, T. Legero, U. Sterr, F. Riehle, M. J. Martin, L. Chen, and J. Ye, "A sub-40-mHz-linewidth laser based on a silicon single-crystal optical cavity," *Nat. Photonics* **6**, 687–692 (2012).
11. F. Kéfélian, H. Jiang, P. Lemonde, and G. Santarelli, "Ultralow-frequency-noise stabilization of a laser by locking to an optical fiber-delay line," *Opt. Lett.* **34**, 914–916 (2009).
12. H. Jiang, F. Kéfélian, P. Lemonde, A. Clairon, and G. Santarelli, "An agile laser with ultra-low frequency noise and high sweep linearity," *Opt. Express* **18**, 3284–3297 (2010).

13. K. H. Wanser, "Fundamental phase noise limit in optical fibres due to temperature fluctuations," *Electron. Lett.* **28**, 53–54 (1992).
14. S. Foster, A. Tikhomirov, and M. Milnes, "Fundamental thermal noise in distributed feedback fiber lasers," *IEEE J. Quantum Electron.* **43**, 378–384 (2007).
15. R. E. Bartolo, A. B. Tveten, and A. Dandridge, "Thermal phase noise measurements in optical fiber interferometers," *IEEE J. Quantum Electron.* **48**, 720–727 (2012).
16. L. Z. Duan, "Intrinsic thermal noise of optical fibres due to mechanical dissipation," *Electron. Lett.* **46**, 1515–1516 (2010).